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EARLY LABOR MARKET TRANSFORMATION UNDER GENERATIVE AI

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Still Waters, Rapid Currents: Early Labor Market Transformation under Generative AI
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ABSTRACT

We study the early labor market impacts of AI chatbots by linking large-scale adoption surveys to administrative labor market records in Denmark. We document rapid currents: most employers in exposed occupations have adopted chatbot initiatives, workers report productivity benefits, and new AI-related tasks are widespread. Yet these currents have not broken the surface: using difference-in-differences, we estimate precise null effects on earnings and recorded hours at both the worker and workplace levels, ruling out effects larger than 2% two years after the launch of ChatGPT. What moves is the structure of work: employers absorb AI through task reorganization—including new tasks in content generation, AI oversight, and AI integration—and adopters transition into higher-paying occupations where AI chatbots are more relevant, though still too few to move average earnings. Technological change reshapes work well before it surfaces in earnings or hours.

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AI chatbots mark the rise of generative artificial intelligence (AI): these tools have seen the fastest worker take-up of any new technology (Bick, Blandin and Deming, 2025), and controlled experiments demonstrate large productivity gains for users (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023). Yet even as generative AI tools have become widespread, substantial disagreement remains about their labor market effects, with forecasts ranging from dramatic displacement within a few years (Amodei, 2025) to modest impacts over a decade (Acemoglu, 2025).¹ This disagreement partly reflects how little evidence exists on how labor markets actually absorb new technologies in their early stages. How does work reorganize at the onset of major technological change?

This paper argues that early labor market transformation under generative AI is best understood as *rapid currents under still waters*: substantial reorganization of work occurs as workplaces integrate AI chatbots and workers take on new AI-related tasks, absorbing change before it surfaces in earnings or recorded hours.

To study this process, we develop a unique data infrastructure linking workplace adoption and work reorganization to administrative labor market outcomes. In collaboration with Statistics Denmark, we conducted a series of large-scale, representative surveys of AI chatbot adoption in 11 highly exposed occupations—where labor market effects are likely to emerge earliest and strongest—linking responses to administrative labor market records.² Our latest survey round, conducted in late 2024, includes responses from 25,000 workers across 7,000 workplaces. These linked survey–register data allow us to examine both recorded labor market outcomes—such as earnings and hours—and adjustments that typically go unrecorded, including changes in job tasks and work organization. Crucially, even within these highly exposed occupations, many otherwise similar workplaces and workers have taken markedly different adoption paths, allowing us to examine how chatbot

¹Anthropic CEO Dario Amodei predicts that AI could eliminate half of all entry-level white-collar jobs, driving unemployment to 10–20% within the next one to five years (Amodei, 2025). In contrast, Acemoglu (2025) predicts that AI will increase annual TFP growth by less than 0.07 percentage points over the next ten years.

²Our list of occupations is accountants, customer support specialists, financial advisors, HR professionals, IT support specialists, journalists, legal professionals, marketing professionals, office clerks, software developers, and teachers.

adoption relates to work reorganization and labor market outcomes.

We start beneath the surface by examining how workers’ use of, and reported benefits from, AI chatbots vary with their employer initiatives. Even in workplaces that neither encourage use nor provide enterprise tools or training, about 40% of workers have used AI chatbots at work, and roughly 7% use them daily. Despite this substantial worker-driven baseline, take-up rates almost double in workplaces with active employer initiatives. Encouraged-use policies, in particular, are associated with greater regular use and higher reported benefits among users, including time savings, quality, creativity, and job satisfaction. Overall, adoption and reported benefits peak in workplaces that combine encouraged use with enterprise chatbots and training: 93% of workers in such settings report having used AI chatbots at work, 28% use them daily, and 19% report saving more than one hour per day—patterns consistent with complementarities between new tools and organizational investments (Bloom, Sadun and Reenen, 2012; Brynjolfsson and Hitt, 2000).

AI chatbots not only affect existing work—they also create new tasks. Even in the absence of employer initiatives, about 8% of chatbot users report taking on entirely new tasks arising from chatbot use, rising to roughly 17% in workplaces with active employer initiatives. These tasks span content *generation* (e.g., drafting and ideation), *oversight* of AI outputs (e.g., quality review and compliance), and *integration* of AI tools into workflows (e.g., developing usage policies). Consistent with this reorganization of work, most chatbot users (85%) report reallocating time savings from AI chatbots to other job tasks. New tasks extend even to workers who have never used AI chatbots, signaling broader workplace transformation. These patterns support task-based theories of how new technologies reinstate labor demand (Acemoglu and Restrepo, 2018a; Autor et al., 2024).

We then rise to the surface, asking whether and how these rapid currents of chatbot adoption and work reorganization register in earnings and recorded hours. Our analysis boils down to a simple question: have chatbot adopters fared differently from comparable non-adopters? To address this question, we link our surveys to administrative records